

Rituraj Singh

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EDUCATION

National Institute of Technology Jamshedpur

2023-2027 (Expected)

B.Tech in Electronics and Communication Engineering / CGPA: 7.34

Central Board of Secondary Education

2021-2023

Class 12th / D.A.V Public School / Percentage: 75.4

PROJECTS

- **Glimpse – A full-stack visual-sharing web app**

React.js, Vite, JavaScript, Node.js, Express.js, MongoDB, Google Gemini API [View Project](#)

- Built a full-stack **visual-sharing platform** using a **React.js + Vite frontend** and a **MERN-based backend**, enabling users to create, explore, save, and interact with image-based content.
- Implemented core features including **Google authentication with guest access control**, **image upload and pin management**, **search with category filtering**, and **commenting**, improving content discovery and engagement.
- Integrated a **Google Gemini API-powered chatbot** to assist users with pin titles, category suggestions, profile improvement ideas, and content planning inside the application.
- Designed a **responsive Pinterest-inspired UI** with Tailwind CSS and React Router, while organizing the codebase with reusable components, API utilities, and modular backend controllers for better scalability.

- **Redis – A lightweight Redis clone built in C++**

C++, TCP Sockets, Multithreading, Ubuntu [View Project](#)

- Designed and developed a **Redis-inspired in-memory key-value store** in **C++**, supporting core commands with millisecond-level response time.
- Implemented a **TCP server using sockets** with RESP protocol parsing to handle multiple concurrent client connections using multithreading and synchronization primitives.
- Added support for **Stream operations** (XADD, XRANGE) with blocking reads and time-based ID generation.
- Built a **Leader-Follower replication model** with replication ID and offset tracking to synchronize data across replicas.

- **Smart Product Pricing Model – A Multimodal AI Solution for Price Prediction**

Python, Pandas, Matplotlib, Word2Vec, Vision Transformer (ViT), LightGBM, PCA [View Project](#)

- Performed extensive **data cleaning and EDA** on 50k+ product records to handle missing values, outliers, and inconsistent units, ensuring high-quality input for modeling.
- Developed a **multimodal model** combining **text (Word2Vec embeddings)** and **image features (Vision Transformer – ViT)** for accurate price prediction.
- Fused visual and textual representations using **PCA** and optimized a **LightGBM regressor**, achieving high prediction accuracy and generalization.
- Applied **log normalization**, **feature scaling**, and iterative evaluation to enhance model robustness and prevent overfitting.

SKILLS

- **Languages:** Java, C++, Python, JavaScript
- **Core Computer Science:** Data Structures and Algorithms, OOP, DBMS, Operating Systems
- **Web Development:** React.js, Node.js, HTML, CSS, Tailwind CSS, REST APIs
- **Tools & Technologies:** Git, Docker, SQL, WSL, Vite
- **Machine Learning:** NumPy, Pandas, Scikit-learn, TensorFlow, NLP, Deep Learning, RNN

ACHIEVEMENTS

- **Competitive Programming-** Solved **600+ problems** across LeetCode, Codeforces, and GeeksforGeeks.
- **Gen AI Study Jam, GDSC-** Completed Generative AI program by **Google Developer Student Clubs**. [View Badge](#)
- **Data Science Certification-** Completed Data Science course covering Python, ML and DL fundamentals. [View Certificate](#)
- **JEE Main-** Secured **AIR 17,543** (General) with **98.48 percentile** among **1.2M+ candidates**.

EXTRACURRICULARS

- **Content Team Member – SECE, NIT Jamshedpur**
 - Played an active role in organizing events and sessions conducted by the Society of Electronics and Communication Engineering (SECE).
- **Volunteer – Sankalp, NIT Jamshedpur**
 - Mentored underprivileged children, providing guidance on academics and career planning.

ONLINE PROFILES

- [LeetCode](#)
- [Codeforces](#)